CSA
Cybersecurity Industry Call for Innovation
Submission Template

Project Title:	**Automated Cyber-Physical Assessment for Smart Power Grids**
Full Company Name:	**Ambient Systems Pte. Ltd.**

Notes to Applicant

1. Follow the slide template sequence strictly, i.e. do not change it
2. Each proposal should only respond to **ONE** Challenge Statement
3. All questions must be answered and preserved, i.e. do not delete any questions
4. Do not exceed 20 slides (excluding title slide, instructions/guidelines, separators, Thank You slide)
5. Convert the slides to pdf before submission
6. Maximum file size for submission: 7MB

CSA CCDF Guidelines

Extract of CSA Cybersecurity Co-Innovation and Development Fund (CCDF) guidelines

- No retrospective applications will be accepted
  - An application is deemed retrospective if the proposed project has already commenced at the time of application
- There must be development of new features, not just integration/configuration of existing products/solutions
- At least 50% of manpower requested for funding must be Singaporean or Singapore PR
- All Singapore registered companies are eligible for the CSA CCDF
  - Overseas firms that are not registered in Singapore will need to partner with a Singapore registered company
- Funding awarded must be used to carry out development activities physically in Singapore
  - Recipients of CCDF funding should use Singapore as a base to own, manage and register all intellectual property developed during the project
- Proposals already funded or considered for funding by other government agencies will not be considered for CCDF
- All claims will be disbursed on a reimbursement basis upon submission of claim document(s) such as salary slips, invoices etc.

Please refer to https://www.csa.gov.sg/programmes/co-innovation-development-fund for the full guidelines

Project Title: Automated Cyber-Physical Assessment for Smart Power Grids
Company Name: Ambient Systems

Project Quad Chart

Summary

Technical Lead: Ivan Damnjanovic, CEO, [redacted]

Illustration
Illustration Work-in-progress

Objectives / Deliverables

Challenge Statement

- CS11: Breach and Attack Simulator (BAS) for an Internet of Things (IoT) connected Power Grid

Objective

- Leverage IoT device-level data to augment cyber security capabilities of extended Power Grid systems

Deliverables

- Please see slide 8

TRL (Technology Readiness Level)

- Start TRL = 2; Projected End TRL = 6

Approach and Innovation*

Proposed Approach

- ➔ Use sample data to generate digital twin of connected devices
- ➔ Create a rule-book for checking device integrity
- ➔ Run a crawler through device network, checking the status of the device using the proprietary rule-book
- ➔ Send data report to user with summary of the results

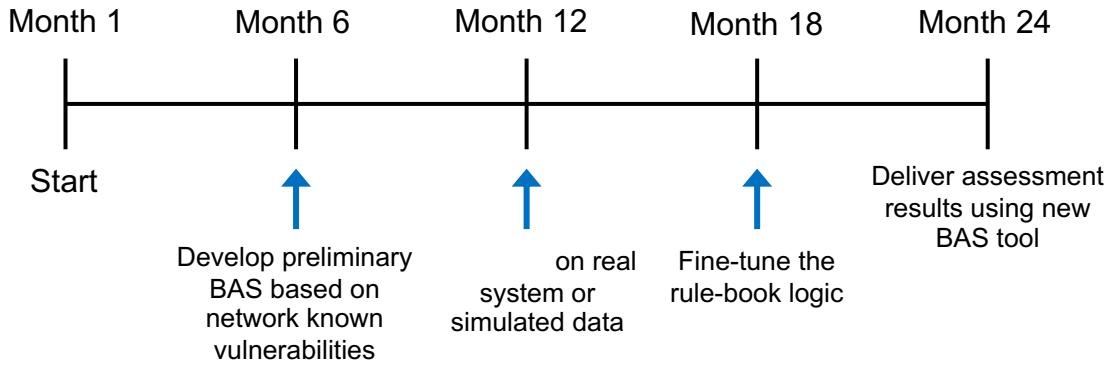
Innovation

- Innovation consists of in the (1) automated process of evaluating device characteristics and (2) rule-book logic on what constitutes a 'compromised' or 'vulnerable' device

***This section is mandatory and will be important for the evaluation**

Funding and Schedule

- **ed**: **<S$ 100,000>**
- **Total Project Cost: <S$ 100,000>**
- **Schedule (Max 24 months)**: 4 significant project milestones



1. Objective and Approach

a) What problem is this proposal trying to solve?

Building a smart utility grid requires addition of new hardware with internet capabilities. This creates a greater stream of data, enabling many use-cases, such as improving energy efficiency or sustainability. However the addition of new devices to the internet network opens up the system to potential attacks. Existing products on the market consider only the computing characteristics of the device to evaluate its integrity. Our solution compliments these data through device-specific parameters when evaluating its integrity.

b) What does the proposed solution do? How will it solve the problem?

Our solution proposes to use device-level physical data in order to evaluate its integrity (in addition to information about network / internet configurations). The type of data we use will depend on the device itself. AMIs or charging stations will likely be equipped with existing sensors that we can use to augment the security assessment of the device. If not, we propose to use device continuous electromagnetic radiation (EMR) profile in order to evaluate its security status. We propose to classify differing EMR profiles using machine learning techniques. This will allow us to create a rule-book of 'approved' device states or operations. If the device EMR deviates from the known (and approved) categories in the rule-book, it would indicate that the device is (1) being used in non-standard way or (2) malfunctioning. Both sets of data are important from the cybersecurity perspective and gives the operator greater visibility into the system. The classification models will be pre-trained and deployed at the edge. Model outputs will be sent to the operators in the format that is insightful for them (e.g., cloud-based dashboard, email notification, message notification, auto-generated report).

1. Objective and Approach

c) Challenge Statement Requirements: State which requirements the proposal will fulfil and which it will not

Delete this slide if responding to CSOC (Open Category)

No.	Requirement	Complied
a	Develop and Maintain a database of vulnerabilities for IoT systems	Yes
b	Identify and flag vulnerabilities detected	Yes
c	Continuously scan, identify and flag new vulnerabilities	Yes
d	Provide remediation and mitigation suggestions on detected vulnerabilities.	Yes
e	Be resilient and fail-secure, isolating faults without affecting the entire grid	Yes
f	Be compatible with all relevant IoT devices in use	Yes

2. Proposed Solution Implementation

a) How does the proposed solution work? How will it be implemented?

The solution architecture consists of several components, including hardware, , and cloud-services (depending on user's information management policy).

In order to deploy our solution, we'll need a controller near the devices to run our monitoring code. This code scans the network end-points to establish a map of devices and their states. The states are based on the network configuration of the devices (i.e., whether they're securely connected to the internet) as well as the physical characteristics of the device (e.g., its electromagnetic radiation (EMR) profile). The controller will have a pre-trained machine learning model running to evaluate the data coming in from the network of devices. The model itself will use classification techniques to categorize the incoming data from devices (e.g., secure, vulnerable, or potentially compromised). The output of the categorization will be sent to the user for evaluation and remediation action. Format of the output can be relayed to a cloud-based dashboard, auto-generated report, email or text notifications, or other modes most accessible to the operators.

2. Proposed Solution Implementation

b) What are the project outputs or upon project completion?

No.	and Specifications
1	Requirements gathering (understand problem statement based on stakeholder discussions)
2	Solution architecture design (based on stakeholder review and discussions)
3	Prototype development (focus on UI/UX design for easy adoption)
4	BAS tool prototype
5	Assessment report
6	Remediation report
7	Documentation / guidelines
8	Adoption / deployment plan
9	Training materials / workshops
10	User manual

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3. Innovation

a) What is the innovation exhibited by the proposed solution?

Key innovation is using physical characteristics of the device in addition to network configuration to establish whether its secure. The method relies on the data available specific to the device, but also proposes a more general approach using electromagnetic radiation (EMR) that can be applied regardless of the device's existing hardware capabilities.

b) Market Comparison on the Proposed Product with Other Products in the Market

No	Features	Your Product	Other Product A	Other Product B
1	Work-in-progress			
2				
3				
4				
5				
6				

4. User Acceptance Test

a) What is the test setup? (include diagram to explain and list any assumptions)

Test setup consists of a smart controller that the operator can plug into their network to run the BAS tool.

We propose running the BAS tool using sampled (or simulated) data first before deploying it onto the network.

The sampled data can be based on statistical sample of the real data (after necessary sanitizing and anonymizing it).

User needs to provide us either a sample data, simulated data (based on the real data), or access to the device network.

Then we simply run the tool using these data and show the user the results.

User needs to tell us how they prefer to see the results.

4. User Acceptance Test

b) What are the test cases and passing criteria ?

No.	Test Case and Objective	Passing Criteria
1		
2		
3		
4		
5		
6		
7		
8		
9		
10		

5. Project Timeline and Milestones

a) Propose the project schedule

No.	Scope of Work, Tasks, Tests, Specification and Deliverables	Months (T + N)	Status & Remarks
1			
2			
3			
Milestone 1: Completion of < >		**T + 6**	
4			
5			
6			
Milestone 2: Completion of < >		**T + 12**	
7			
8			
9			
Milestone 3: Completion of < >		**T + 18**	
10			
11			
12			
Milestone 4: Completion of < >		**T + 24**	

6. Project

Category	Description	Total Amount (S$)
A. Expenditure on in-company Manpower (EOM) (Will be reimbursed via validation of **salary** documents)	Example Software Engineer: $30,000 (6 mths x $5,000) Hardware Engineer: $30,000 (6 mths x $5,000)	
B. Expenditure on Third Party expenses such as purchases of Equipment/Software/hardware/Professional Services (Will be reimbursed via validation of **invoice** documents)	Example Web Application Firewall: $4,000 Network Firewall: $3,000 Hardware Security Module (HSM): $7,000	
C. Other Operating Expenses (OOE)		
Total ed (A+B+C)		
(e.g. manpower, equipment) that the Applicant Company and/or collaborators (if any) will self-fund	Example Consultant: $30,000 (6 mths x $5,000) Project Manager: $30,000 (6 mths x $5,000) New Logic Analyser: $5,000	
Total Self-Funding (D)		
Total Project Cost (A+B+C+D)		

7. Project Impact

a) What is the potential for the solution to be developed into a commercial product for industry-wide adoption?

b) What is the estimated budgetary pricing of the solution after the project has been completed successfully?

No	Description of components	One-Time-Cost/CAPEX (S$)	Annual Recurring Cost/OPEX (S$)	Price/Discount for current End User
1.				
2.				
3.				

c) What is the estimated international addressable market size for the solution?

8. Company Background

No	Category	Sub-category	Value
1	**Full company name**		Ambient Systems Pte. Ltd.
2	**Company global headquarter**	Country	Singapore
3	**Years of establishment**	Global	2020
		Singapore	2020
4	**Number of employees**	Global	**4 full-time + 2 part time**
		Singapore	**4 full-time + 2 part time**
5	**Core business activities**	Key products/services	**AI solutions for building management**
		Existing clientele	**Deep tech spin-off from National University of Singapore's Graduate Research and Innovation programme (GRIP)**

Shareholding Structure

No	Share Holder Name	Percentage of Holding	
1	Ivan Damnjanovic	100%	Canadian
2	NUS -- GRADUATE INVESTMENT PRIVATE LIMITED		N/A
3			
4			

9. Project Team

a) List all team members involved in this project

S/N	Name	Job scope in project (list at least 2-3 points)	in which Country	Company	Years of experience	Funding Needed (Y/N)
1						
2						
3						
4						
5						
6						
7						

b) Does the core team reside in Singapore and project will be developed in Singapore? (Yes/No)

10. Others

a) Any other considerations that will bring merit to the proposed solution

b) Does the proposed solution require longer term Research and Development (R&D)?

Thank You